

Cloud LearnX

Mathematics

Trigonometry — Practice Worksheet

Student: Kalan

Date: 07 August 2026

Topic: Trigonometry

Instructions:

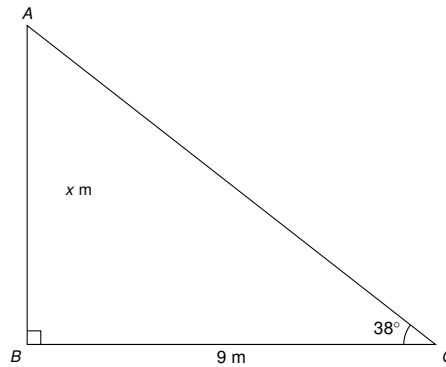
- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

Answer ALL ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1** A space telescope's support strut AC is fixed to a vertical antenna mast AB . The strut makes an angle of 38° with the ground at C , and the foot of the mast B is 9 m from the base of the strut C , as shown in the diagram below.



Calculate the length AB , marked x on the diagram, giving your answer correct to 3 significant figures.

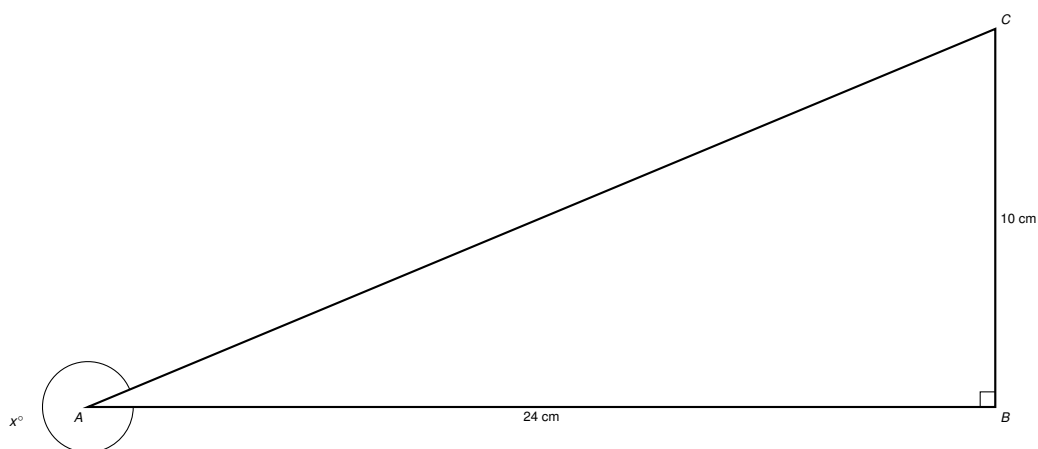
..... m

(3)

(Total for Question 1 is 3 marks)

2 An archivist is examining an old wooden book stand, which is designed as a right-angled triangular support, ABC , used to prop open a large historical manuscript.

The base AB of the stand is 24 cm long, and the vertical support BC is 10 cm long, with the right angle at B .



Work out the size of angle x , the angle between the base AB and the sloping edge AC . Give your answer correct to 1 decimal place.

$x = \dots\dots\dots^\circ$
(3)

(Total for Question 2 is 3 marks)

3 A space agency's engineering lab has a diagonal support strut that braces a satellite testing platform against a wall.

The strut makes an angle of 37° with the horizontal floor. The foot of the strut is fixed to the floor at a horizontal distance of 2.4 m from the wall.

Calculate the length of the support strut.

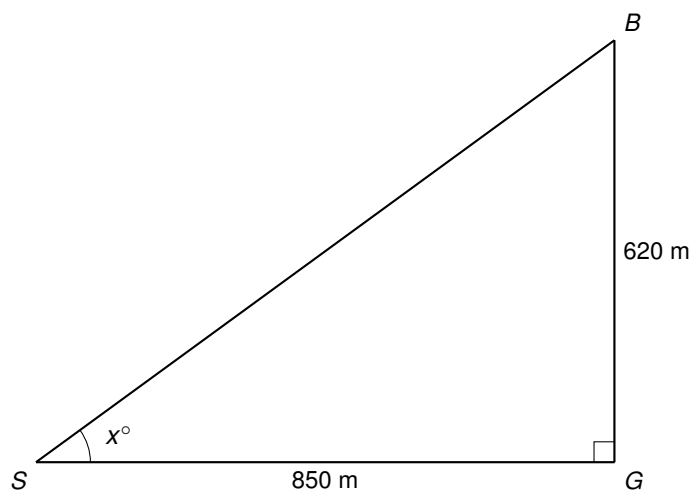
..... m

(3)

(Total for Question 3 is 3 marks)

4 A space agency uses a satellite tracking station to observe a rocket launch. The tracking station is at ground level, at point S .

A weather balloon released to monitor atmospheric conditions is directly above a point G on the ground, where G is 850 m horizontally from the tracking station S . The balloon, at point B , is at a height of 620 m above the ground.



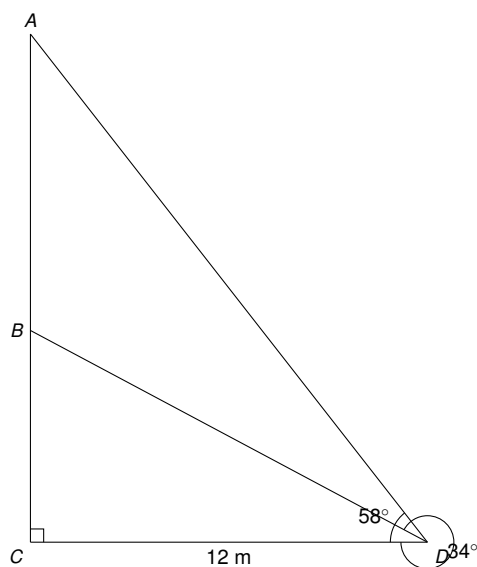
Calculate the angle of elevation, marked x° on the diagram, of the balloon B from the tracking station S .

$$x = \dots\dots\dots^\circ$$

(4)

(Total for Question 4 is 4 marks)

- 5 An astronomer sets up a telescope tracking tower, modelled by the two right-angled triangles ABD and BCD shown below. ABC is a straight vertical line, and D is a fixed observation post on the ground.



C is the base of the tower, D is the observation post on the ground, and $CD = 12$ m. The angle of elevation of A (the top of the tower) from D is 58° . B is a point on the tower, and the angle of elevation of B from D is 34° .

- (a) Calculate the height AC of the tower.

$$AC = \dots\dots\dots \text{ m}$$

(3)

- (b) Calculate the length AB , the distance between the observation platform B and the top of the tower A .

$$AB = \dots\dots\dots \text{ m}$$

(2)

(Total for Question 5 is 5 marks)

6 A historian is analysing the trajectory data of an early satellite launch. As part of the calculation, she needs to solve the equation

$$\sin \theta = 0.629$$

for an angle θ that lies between 0° and 180° .

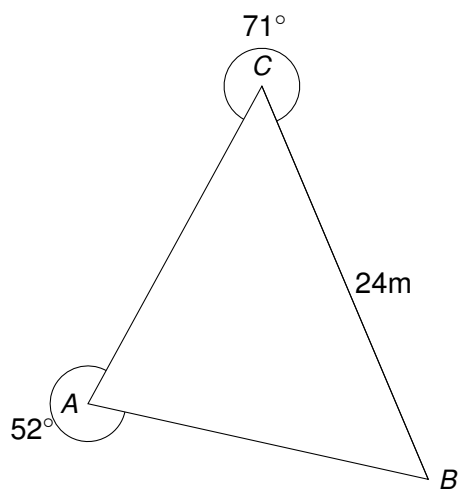
Find **both** possible values of θ in this range, giving each answer correct to 1 decimal place.

(3)

(Total for Question 6 is 3 marks)

7 A historian is studying an old survey map of a triangular plot of land, ABC , marked out around the ruins of a Roman observatory once used to track celestial events.

The surveyor recorded that $BC = 24$ m, angle $BAC = 52^\circ$, and angle $ACB = 71^\circ$.



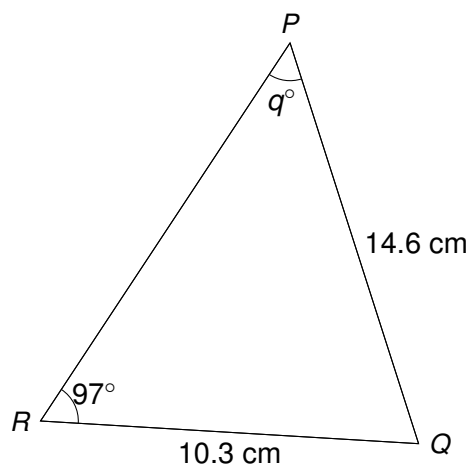
Calculate the length of AB . Give your answer correct to 3 significant figures.

..... m

(4)

(Total for Question 7 is 4 marks)

- 8 A historian is studying an old triangular star map recovered from an archive documenting early space observations. The triangular plot connects three observation points, P , Q and R , marked on the map. $PQ = 14.6$ cm, $QR = 10.3$ cm, and the angle at R (angle PRQ) is 97° .



Work out the size of angle QPR , marked q° on the diagram. Give your answer correct to 1 decimal place.

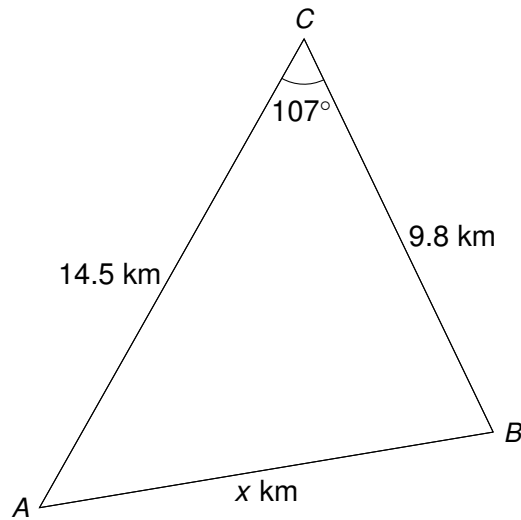
$$q = \dots\dots\dots^\circ$$

(4)

(Total for Question 8 is 4 marks)

- 9 A space agency is tracking two ground stations, A and B , that both observe a weather balloon at point C as it rises through the atmosphere.

The distance from station A to the balloon is $AC = 14.5$ km, and the distance from station B to the balloon is $BC = 9.8$ km. The angle between these two lines of sight, $\angle ACB$, is 107° .



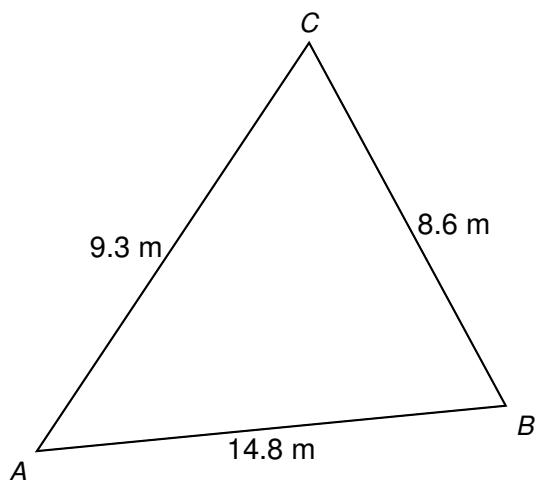
Calculate the distance AB between the two ground stations. Give your answer correct to 3 significant figures.

$AB = \dots\dots\dots$ km

(4)

(Total for Question 9 is 4 marks)

- 10** A historian is studying the triangular layout of an ancient observation site, thought to have been used for tracking celestial events. Archaeologists have measured the distances between the three marker stones, A , B and C , as shown in the diagram.



$AB = 14.8\text{ m}$, $BC = 8.6\text{ m}$ and $AC = 9.3\text{ m}$.

Calculate the size of the largest angle in the triangle.

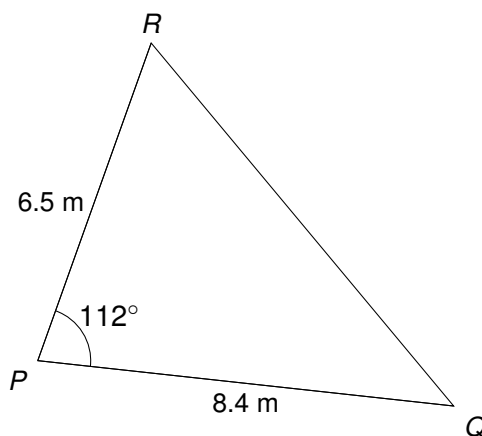
$angle = \dots\dots\dots^\circ$

(4)

(Total for Question 10 is 4 marks)

- 11** A history museum is designing a triangular display plinth to hold a scale model of a triangular sail from an early solar-sail spacecraft concept.

The plinth is triangular in shape, with $PQ = 8.4$ m and $PR = 6.5$ m, and the angle between these two sides, angle QPR , is 112° .



Calculate the area of the triangular plinth PQR . Give your answer correct to 3 significant figures.

..... m^2
(3)

(Total for Question 11 is 3 marks)

12 Three remote research stations, P , Q and R , are used to track the trajectory of a high-altitude balloon launched as part of a stratospheric observation programme.

Station Q is 14.8 km from station P . Station R is 21.3 km from station Q .

The angle between the lines of sight PQ and QR , measured at station Q , is $A = 137^\circ$.

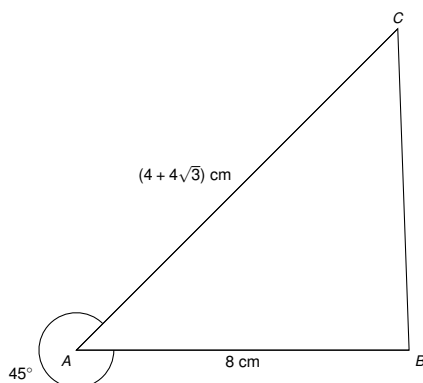
Calculate the distance PR between stations P and R , giving your answer to 3 significant figures.

..... km

(5)

(Total for Question 12 is 5 marks)

- 13** A museum is designing a triangular display case to hold a scale model of a meteorite fragment. The triangular base ABC of the case has $AB = 8$ cm, $AC = (4 + 4\sqrt{3})$ cm, and angle $BAC = 45^\circ$, as shown in the diagram (not drawn to scale).



Show that the exact area of triangle ABC can be written in the form $(a + b\sqrt{3})$ cm², where a and b are integers to be found.

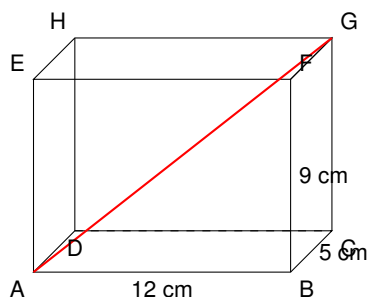
(5)

(Total for Question 13 is 5 marks)

- 14** A space agency stores a satellite instrument inside a cuboid-shaped protective casing $ABCDEFGH$, as shown in the diagram.

The casing has length $AB = 12$ cm, width $BC = 5$ cm, and height $CG = 9$ cm.

During transport, the instrument must be packed with a rigid support rod placed along the diagonal AG , from one bottom corner to the opposite top corner of the casing.



Calculate the length of the diagonal AG .

Give your answer correct to 3 significant figures.

..... cm

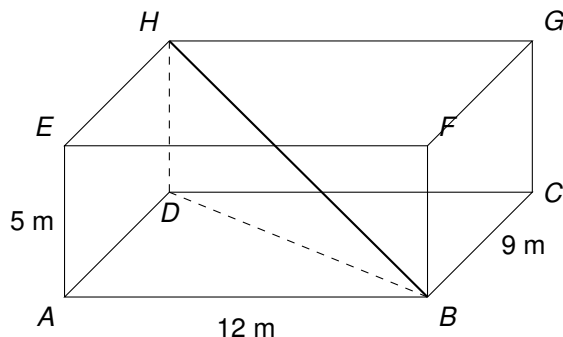
(5)

(Total for Question 14 is 5 marks)

- 15** A space observatory has a telescope mounting platform designed as a cuboid base $ABCD-EFGH$, used to keep a telescope perfectly stable while tracking objects in orbit.

The base $ABCD$ is horizontal, with $AB = 12$ m and $BC = 9$ m. The vertical support struts have height $AE = BF = CG = DH = 5$ m.

A control cable runs in a straight line from H down to the corner B , as shown in the diagram.



- (a) Calculate the length of the cable HB .
Give your answer correct to 3 significant figures.

..... m

(3)

- (b) Calculate the angle that the cable HB makes with the horizontal base plane $ABCD$.
Give your answer correct to 1 decimal place.

$\theta =$ $^{\circ}$

(3)

(Total for Question 15 is 6 marks)

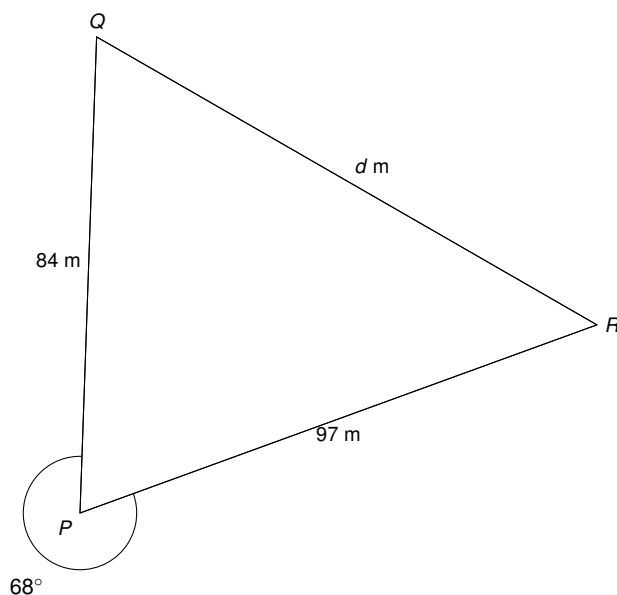
16 A team of planetary geologists is surveying a triangular rock formation, believed to be an ancient meteor impact site, from a fixed base station P on flat ground.

Two distinctive rock outcrops, Q and R , are visible from P .

The distance from P to Q is 84 m.

The distance from P to R is 97 m.

The angle $QPR = 68^\circ$.



The geologists need to know the distance QR between the two outcrops, and the total surface area of the triangular region PQR enclosed by the three survey points, in order to plan a robotic sampling route.

(a) Calculate the distance QR , giving your answer to 3 significant figures.

(b) Calculate the area of the triangular region PQR , giving your answer to 3 significant figures.

(c) The geologists plan to deploy a robotic rover that can safely traverse at most 0.9 hectares (9000 m^2) of terrain per survey. Determine, showing full working, whether the triangular region PQR falls within this limit.

(7)

(Total for Question 16 is 7 marks)

TOTAL FOR PAPER IS ?? MARKS